RACHEL N. SLAYBAUGH

DCVC ⋄ 270 University Ave ⋄ Palo Alto, CA 94301
rachel.slaybaugh@dcvc.com ⋄ (415) · 841 · 3935

EXPERIENCE & ACCOMPLISHMENTS

Data Collective Venture Capital (DCVC) Feb. 2022 – present
Partner *Palo Alto, CA*

- Source, select, fund, and support deep tech companies in the climate space
- Board member for Radiant (independent, 04/22-present), Fervo Energy (04/24-present), Fourth Power (11/23-present), Equilibrium Energy (12/23-present), and Unspun (05/24-present); Board observer for Brimstone (03/22-present), Fervo (08/22-04/22), and Verdigris (08/23-present)

Lawrence Berkeley National Laboratory Jan. 2021 – Dec. 2021
Cyclotron Road Division Director *Berkeley, CA*

- Sourced and selected ∼10 hard tech innovators per year for a 2-year fellowship program to turn their technology concept into a product with a positive societal impact (∼$6M/year)
- Fellows embed at LBNL where we support their technical development and collaborations
- Responsible for outcomes, safety, and reporting to Department of Energy and LBNL leadership

Advanced Research Projects Agency – Energy Jan. 2017 – Oct. 2020
Program Director *Washington, DC*

- Created programs and selected teams developing technologies for advanced nuclear fission reactors, $95M across 28 teams: MEITNER, the Nuclear OPEN+ cohort, LISE, and GEMINA
- Provided deep technical and business guidance while managing: the nuclear teams; the TERRA and ROOTS Programs, supporting research for sensing and data analytics for above- and below-ground plant outcomes (∼$90M across 18 teams); and the FOCUS Program, supporting research for solar technologies that combine photovoltaic and concentrated solar power technologies (∼$12M across 4 teams)
- Supported program creation in other energy areas through workshops, brainstorming, and feedback

University of California, Berkeley Jan. 2014 – Dec. 2021
Associate Professor of Nuclear Engineering *Berkeley, CA*

- Founded the Nuclear Innovation Bootcamp, which brings diverse students from around the world to learn skills essential to innovation in nuclear energy
- Developed numerical methods for neutral particle transport with an emphasis on supercomputing and advanced architectures; applications in reactor design, shielding, and nonproliferation
- Also mentored PhDs in optimization, thermal fluids, and cryptography and anomaly detection
- Published 25 journal articles, 44 refereed conference proceedings, 3 technical reports, 2 book chapters, 5 open source pieces of software, and 2 policy pieces
- Graduated 9 PhD and 4 MS students; research adviser for 1 assistant project scientist, 1 postdoctoral scholar, 1 visiting scholar, and 15 undergraduate students
- Won >$2.5M as principal investigator (PI) and >$26M as co-PI for numerical methods research
- Created and supported a course in which Berkeley students guide second graders at under-served elementary schools in Oakland in hands on science experiments

Bettis Laboratory Mar. 2012 – Aug. 2014
Senior Engineer in the Shield Design and Development group *West Mifflin, PA*

- Implemented and developed several methods for variance reduction in Monte Carlo neutral particle transport that made problems perviously too hard to solve with this high-accuracy tool solvable
- Qualified these methods and software for use in shield design to dramatically reduce time and improve accuracy in design calculations

University of Wisconsin–Madison Sept. 2006 – Nov. 2011
Research Assistant / Rickover Fellow *Madison, WI*

- Dissertation: "Acceleration Methods for Massively Parallel Deterministic Transport" where I added 3 new methods to Denovo, software from Oak Ridge National Laboratory, that are still used
- Developed two Monte Carlo source sampling methods for arbitrarily shaped plasma sources

Penn State Breazeale Reactor Aug. 2003 – Apr. 2006
Reactor Operator *University Park, PA*

- NRC licensed Reactor Operator for TRIGA Mark III reactor
- Analyzed core burn-up anomaly; calibrated gamma irradiation facilities

EDUCATION

Ph.D.	**University of Wisconsin–Madison**, Nuclear Engineering and Engineering Physics with a certificate in Energy Analysis and Policy	2011
M.S.	**University of Wisconsin–Madison**, Nuclear Engineering and Engineering Physics	2008
B.S.	**Pennsylvania State University**, Nuclear Engineering	2006

LEADERSHIP & SERVICE

Non-Profit Boards and Leadership

LabStart Advisor	2022-prsent
National Renewable Energy Laboratory Investor Advisory Board (Chair 2024)	2022-present
National Academies of Science member of the Committee on Laying the Foundations for New and Advanced Nuclear Reactors in the United States	2020-2022
Activate Leadership Council	2021-present
Biden-Harris Transition Team, Department of Energy Agency Review Team	2020
Good Energy Collective, Founding Board Chair and Board Member	2020-present
Pennsylvania State University, Nuclear Alumni Advisory Council	2020-2021
Nuclear Science and Engineering Editorial Advisory Board	2020-2023
University of Michigan, NERS Department Advisory Board	2019-2021
Nuclear Energy Advisory Committee (a U.S. FACA), Appointed Member	2016-2017

Software and Computing

Berkeley Institute for Data Science	Senior Fellow; Advisory Board Member 2014-2021
Berkeley Research Computing	User Advisory Group 2014-2021
The Hacker Within	UCB Faculty Advisor 2014-2017; UW co-founder 2009

American Nuclear Society

Math and Comp. Division	Chair rotation 2016-2019, Exec. Comm. 2013-2016
Rad. Protection and Shielding Div.	Exec. Comm. 2015-2018
other Past Chair / Vice Chair	NEED Comm, Professional Divisions Comm, Student Sections Comm, Professional Women in ANS

Reviewer for US DOE ARPA-E, Fusion Energy Sciences, Technology Commercialization Fund, Canadian Innovation Fund, and a number of journals and technical conferences